NCP shortest words: independent J3 review

TASK-20261031-f9c13a · 1 November 2026

Derived result. For finite B, e in {+1,-1}, and integer coordinate vector a, the minimum is the sum of the absolute coordinate values plus the negation penalty shown below. The proof counts signed translation contributions and negations separately, then attains both lower bounds.

Shortest words from positive block P and negative block N

e = +1, all a_i ≥ 0	P	length = Σ |a_i|
e = -1	N ⟶ ν ⟶ P	length = Σ |a_i| + 1
e = +1, some a_i < 0	ν ⟶ N ⟶ ν ⟶ P	length = Σ |a_i| + 2

Each N translation has one negation to its right; each P translation has zero. TASK-20261031-f9c13a

Frozen blind evidence. The reasoning and every word with B≤2 and length≤5 were committed before any producer or plan read. Initial commit: 8b53f4c94b2b06f6cfdebfcd89b3ed2fd6ef6d9d. Its timestamp is 2026-11-01 03:53:46 UTC. The input came only from the pinned blind-input.yaml in PR 2847. The raw enumeration contains 433 words and 97 distinct states. All 146 sign/vector comparisons with sum |a_i|≤5 agree: 97 minima observed, 49 beyond the length window, zero mismatches.

a	e	Minimum	Enumeration controls
empty, B=0	+1	0	budgets 0 yes; 1 yes
empty, B=0	−1	1	budgets 0 no; 1 yes
(-1,2)	+1	5	budgets 4 no; 5 yes
(-1,2)	−1	4	budgets 4 yes; 5 yes
(2,-2)	+1	6	budgets 4 no; 5 no
(2,-2)	−1	5	budgets 4 no; 5 yes

Post-freeze comparison. Producer revision f4330e1a29feb8fdea9f1eab7131ae5cde2e2b6f uses the same integer counts, right-of-translation convention, sign assumption, penalty and at-most-length iff. Source correspondence is faithful at this scope. The length-6 mixed witness follows from the proof, beyond the search.

Workflow record. The blind phase respected all exclusions. The later comparison read four protected producer files as allowed by the handoff after the first commit. The pinned plan lacks the checker's prospective phased_release field, so the full-session union of reads fails its independence rule. Both phases and the failure are retained. No sibling report or producer receipt was read. This reviewer performed no Lean build.

Verdict and next action. J3 mathematical verdict: holds. Fresh machine status: not_verified. Full formal-workflow acceptance remains inconclusive; the Coordinator must compose the own-joint reports and retain the phase-contract failure. No point-action or algebra-kernel verdict is supplied here.

Reproduce and inspect. The task directory contains enumerator.py, check_formula.py, raw-words.csv, controls.json and initial-freeze.json. review.md gives the full argument and immutable source links. raw/source-custody.json binds all source snapshots. raw/independence-check.json preserves the actual checker result.